ServiceNow-Based Employee Laptop Request Application

Project Documentation

Platform: ServiceNow | Application: Project Application

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1. Introduction

1.1 Project Title

ServiceNow-Based Employee Laptop Request Application

1.2 Project Summary

This project implements a centralized, controlled process for employees to request work laptops. The application uses ServiceNow platform components to capture request information, create a request record, initiate manager approval, apply duplicate-request validation, and communicate the approval outcome to the requester.

1.3 Project Type

Individual project implemented within a ServiceNow instance.

2. Project Overview

2.1 Purpose

The primary business objective of this project is to implement a centralized and controlled system for managing employee laptop requests in a new IT organization.

2.2 Business Goals

- Provide a single platform for employees to request laptops.
- Differentiate laptop requests based on usage type (Basic or Advanced).
- Prevent duplicate laptop requests by the same employee within a year.
- Improve visibility and tracking for IT and management teams.
- Ensure proper approval status tracking and accountability.
- Replace manual processes such as emails and verbal requests with a standardized digital workflow.

2.3 Main Features

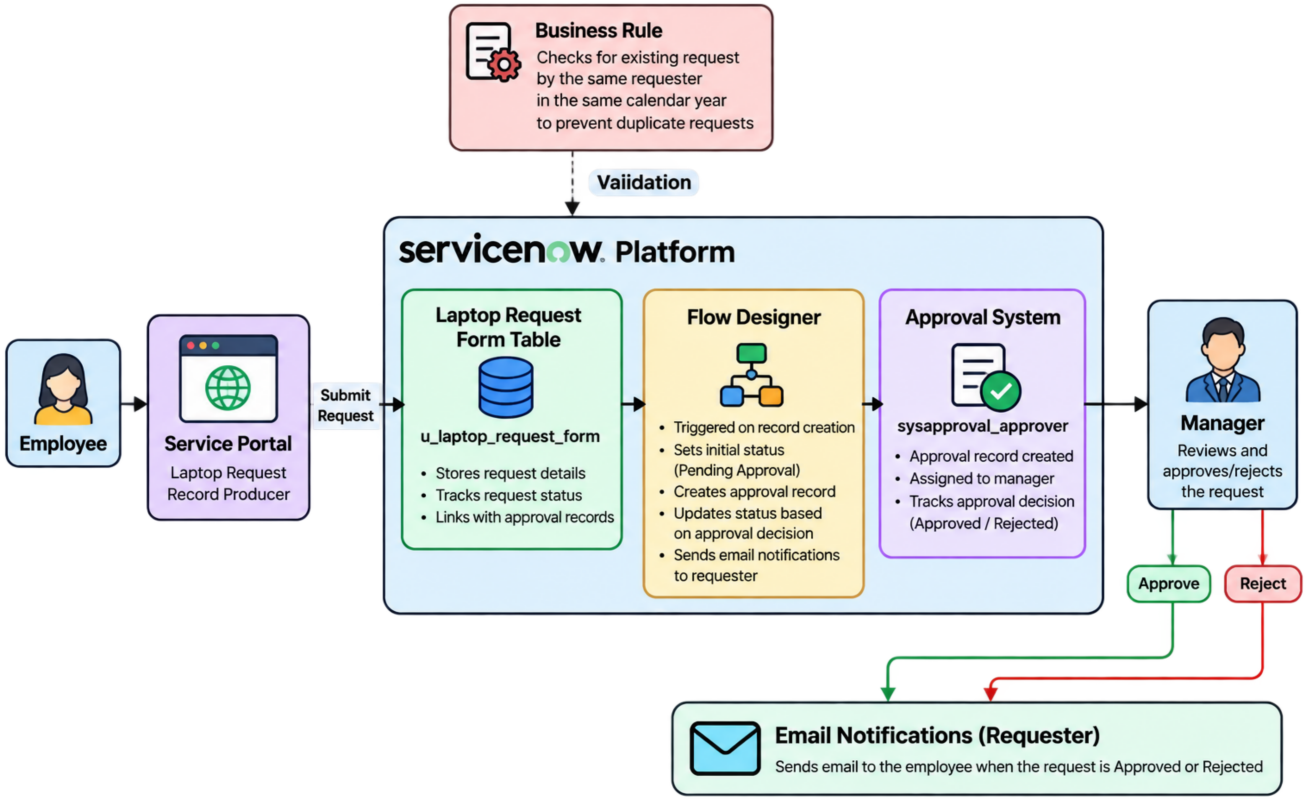
- Employee laptop request submission through the ServiceNow Service Portal using a Record Producer.
- System-generated request number and stored request details.
- Request categories for Basic and Advanced laptop usage.
- Manager selection on the request form and approval records created through the ServiceNow approval process.
- Business Rule to prevent the same requester from submitting more than one request in the same calendar year.
- Flow Designer workflow with approval and rejection branches.
- Email notifications to the requester after approval or rejection.
- Role/group and ACL configuration for access control.

3. Architecture

The application is built on the ServiceNow platform. It does not require a separate frontend framework, backend server, or external database for its core request and approval workflow.

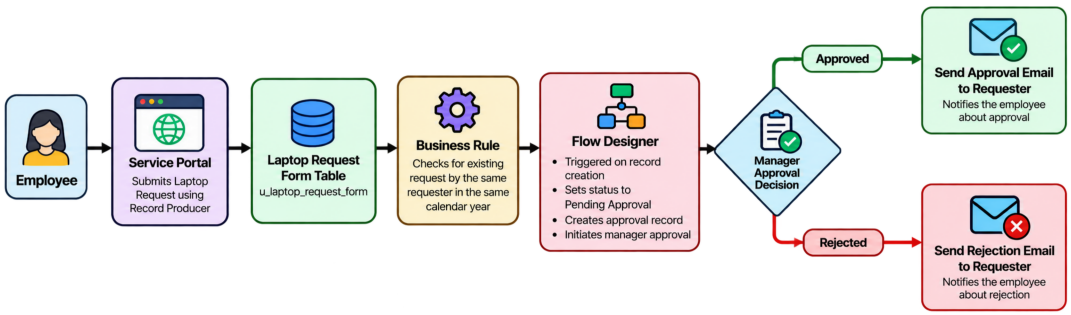
Layer / Component	Implementation	Purpose
User interface	ServiceNow Service Portal and Record Producer	Allows employees to submit a laptop request.
Application	Custom application: Project Application	Contains the project configuration and related components.
Request data	Laptop Request Form table (u_laptop_request_form)	Stores request details, requester, manager, type, reason, status and generated number.
Workflow	Flow Designer	Starts when a Laptop Request Form record is created and runs the approval process.
Approval data	sysapproval_approver	Stores manager approval records and decisions.
Validation	Business Rules	Enforces the annual duplicate-request rule.

Access control	ServiceNow roles, groups and ACLs	Supports controlled access for employee and manager roles.
Notifications	Flow Designer Send Email actions	Sends requester emails for approval and rejection outcomes.
Platform data storage	ServiceNow platform database	Persists application and request records.



3.1 Request Flow

1. Employee opens the Service Portal and submits the Laptop Request Record Producer.
2. ServiceNow creates a record in u_laptop_request_form and generates the request number.
3. A Business Rule checks whether the requester already has a request in the same calendar year.
4. The active Laptop request flow starts when the request record is created.
5. The flow sets the request status to Pending Approval.
6. The Ask For Approval action initiates the approval process.
7. The flow checks the approval outcome and follows the approved or rejected branch.
8. The corresponding Send Email action notifies the requester of the outcome.



4. Setup Instructions

4.1 Prerequisites

- Access to a ServiceNow instance with permissions to configure applications, tables, Record Producers, Business Rules, ACLs and Flow Designer.
- A configured ServiceNow application scope named Project Application.
- Appropriate access to configure users, groups, roles and approval records.

4.2 Configuration Components

1. Create or open the custom application Project Application.
2. Configure the Laptop Request Form table (u_laptop_request_form) and its fields.
3. Create the Laptop Request Record Producer and map its variables to the table fields.
4. Configure the Business Rule that blocks duplicate requests by the same requester within a calendar year.
5. Configure the employee and manager groups/roles and relevant ACLs.
6. Create and activate the Laptop request flow in Flow Designer.
7. Configure requester email actions for approved and rejected outcomes.

4.3 Authentication

The application uses standard ServiceNow authentication. No separate custom login system is part of the implementation.

5. ServiceNow Application Structure

As this is a ServiceNow application, a conventional source-code folder structure does not apply. The implementation is organized as platform records and configuration components.

Component	Technical name / details
Custom application	Project Application
Request table	Laptop Request Form (u_laptop_request_form)
Approval table	sysapproval_approver
Request entry	Laptop Request Record Producer in the Service Portal / Employee Service Catalog
Workflow	Laptop request flow in Flow Designer
Validation	Business Rule for same-requester, same-calendar-year duplicate prevention
Access control	IT Employees and IT Managers groups; it_employee and it_manager roles; ACLs
Notifications	Send Email actions in the approved and rejected flow branches

6. Running the Application

1. Sign in to the ServiceNow instance using the standard login.
2. Open the Service Portal and locate the Laptop Request Record Producer.
3. Enter the employee details, email address, requested date, short description, laptop type, reason and manager.
4. Submit the request. The request number is generated by the system.
5. The created record triggers the Laptop request flow and the initial status is set to Pending Approval.
6. The manager reviews the approval record in ServiceNow and records an approval decision.
7. The flow follows the approved or rejected branch and sends the corresponding requester email.

7. API Documentation

No custom REST API endpoints are identified in the current project scope. The core implementation uses native ServiceNow components, including the Record Producer, custom table, Business Rules, approval records and Flow Designer.

External API integration: Not applicable to the current implementation.

8. Authentication and Access Control

8.1 Authentication

Users access the application through standard ServiceNow authentication.

8.2 Roles and Groups

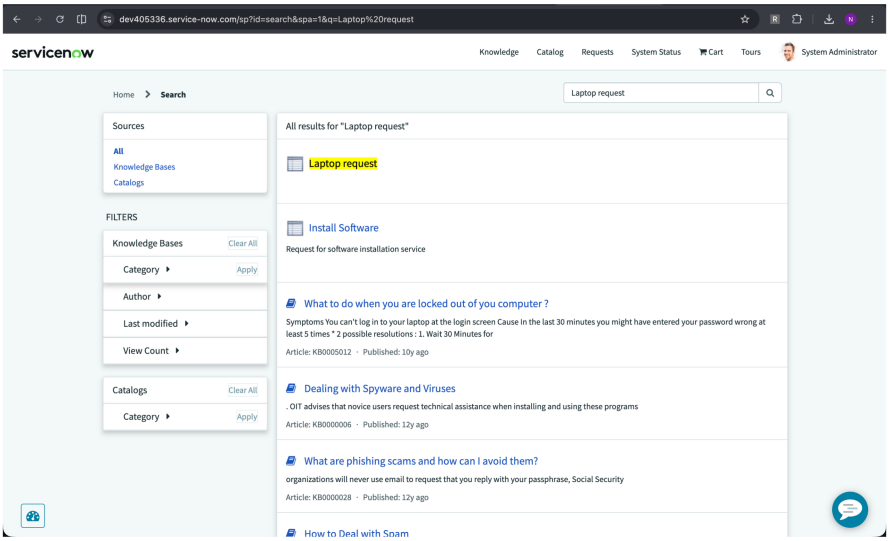
Group	Role	Purpose
IT Employees	it_employee	Represents the employee group configured for the application.
IT Managers	it_manager	Represents the manager group configured for approval responsibilities.

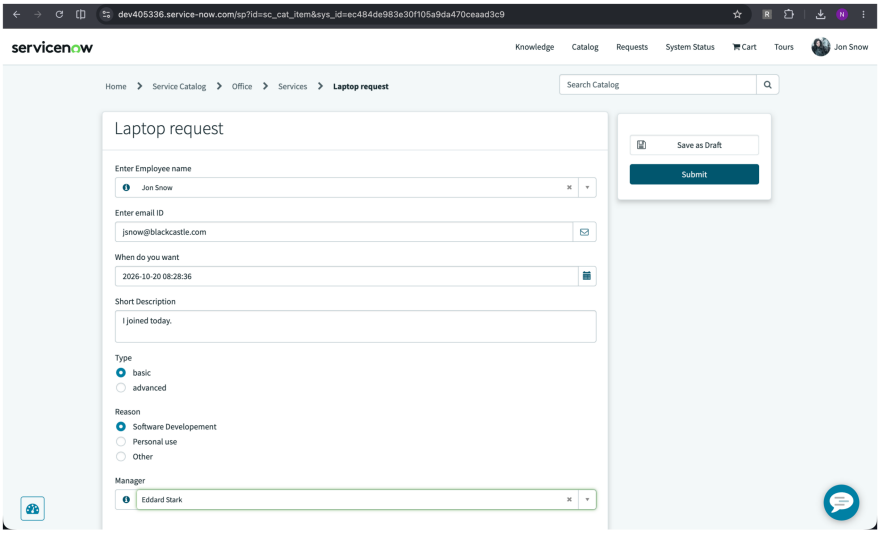
Access is controlled through ServiceNow roles and ACLs. Exact table-level and field-level permissions should be described only where they are confirmed in the instance configuration.

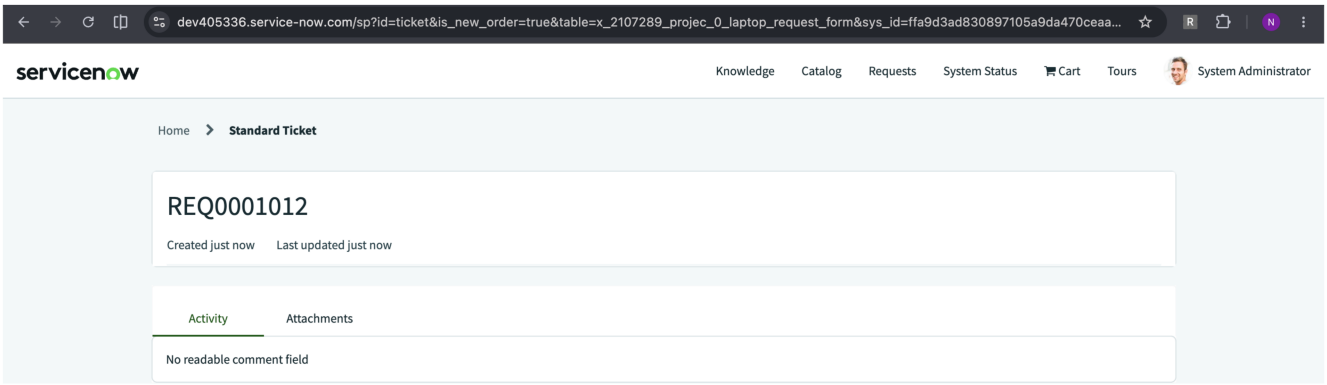
9. User Interface

The employee-facing interface is a ServiceNow Service Portal Record Producer. The form collects the information needed to create and route a laptop request.

1. Service Portal laptop request form







10. Testing

10.1 Testing Approach

User Acceptance Testing (UAT) was performed using defined scenarios and recorded outcomes. Evidence includes test cases, Pass/Fail results, screenshots and a formal UAT summary/report.

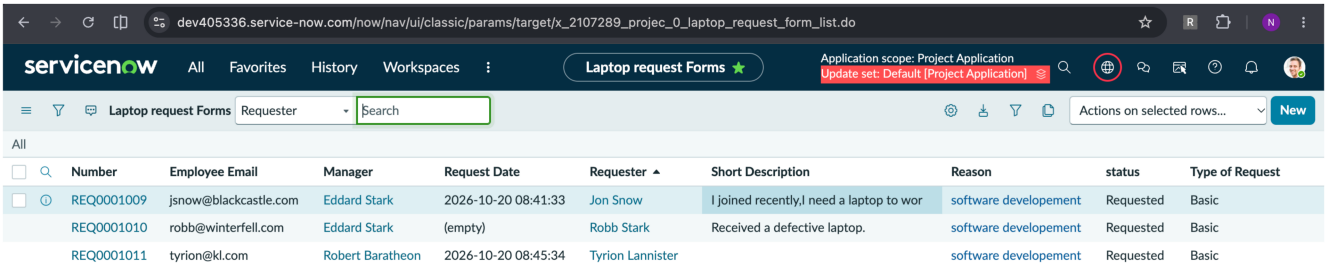
10.2 Confirmed Test Outcomes

Test scenario	Recorded outcome
Duplicate request within the same calendar year	The Business Rule is intended to prevent a duplicate request from the same requester in the same calendar year.
Duplicate request across different calendar years (UAT-17)	Pass: a new request was accepted when the same requester's existing request belonged to a different calendar year.
Initial status before approval (UAT-20)	Pass: the initial request status was Pending Approval.
Email notification after rejection (UAT-21)	Pass: the requester received an email reflecting the rejection.
Requester email after approval	Confirmed by the project owner as working.
Application deployment	Confirmed by the project owner as working in the target ServiceNow instance.

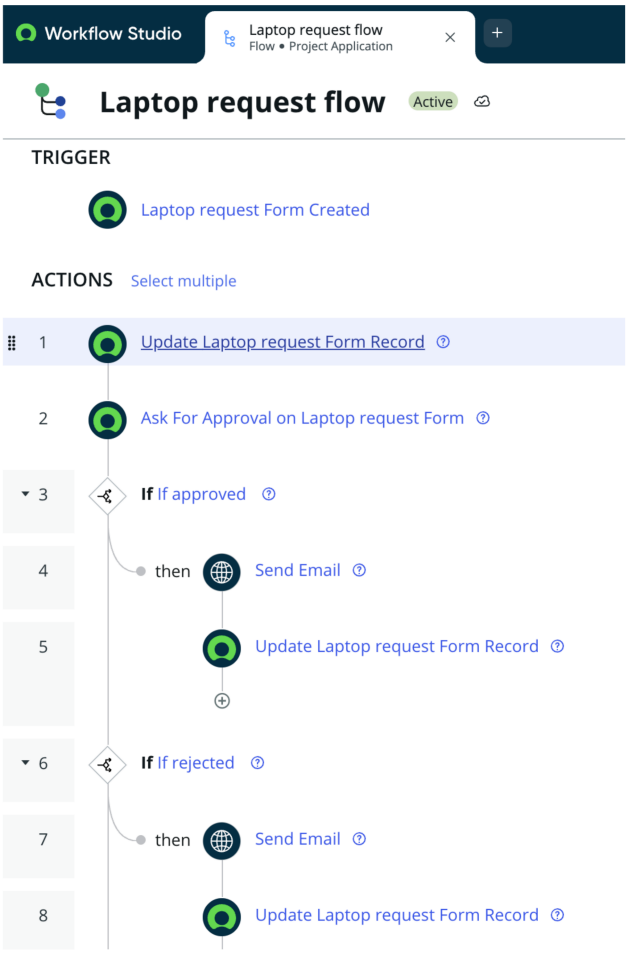
The project's recorded UAT summary reported 20 Pass, 0 Fail and 0 Not Tested across the 20 cases recorded at that point. UAT-21 was subsequently recorded as Pass. Use the final formal UAT report as the authoritative source if its totals differ.

11. Screenshots or Demo

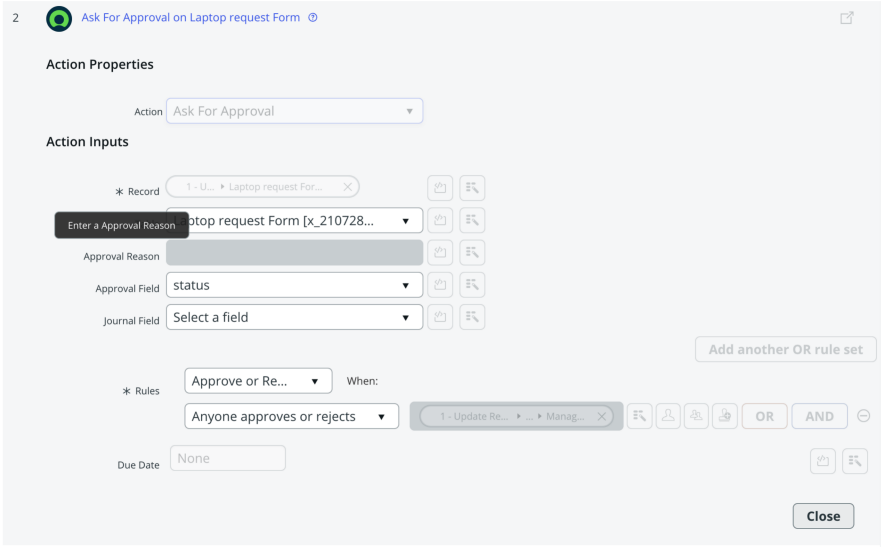
1. Laptop Request Form table showing saved records



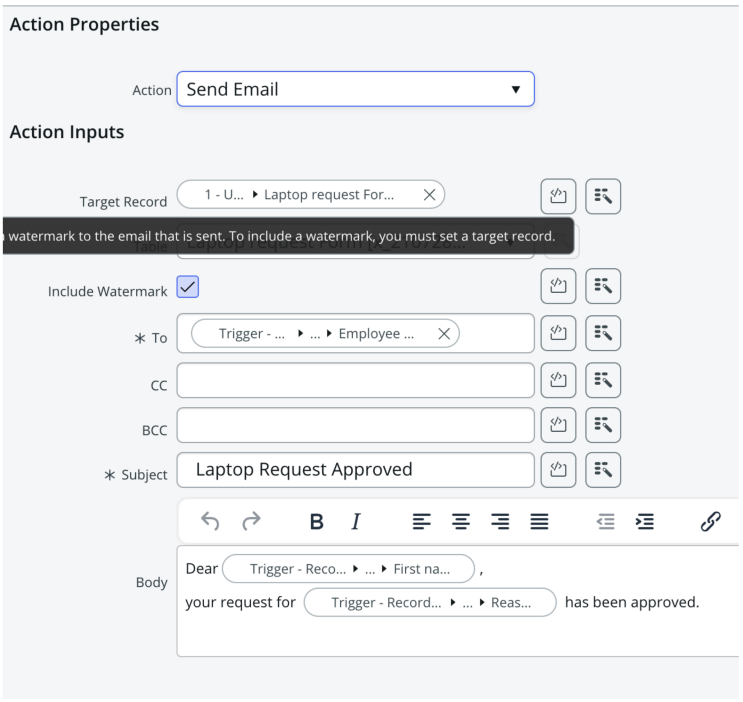
2. Flow Designer workflow

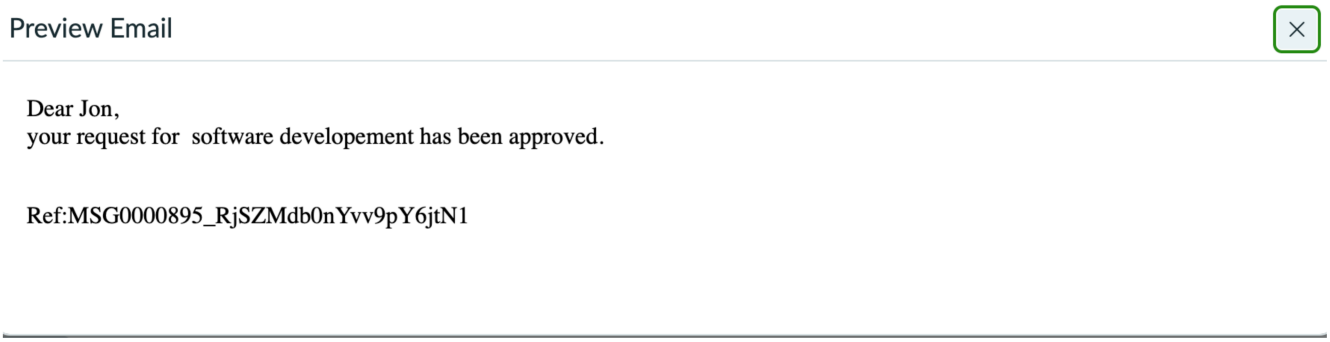


3. Approval record assigned to a manager

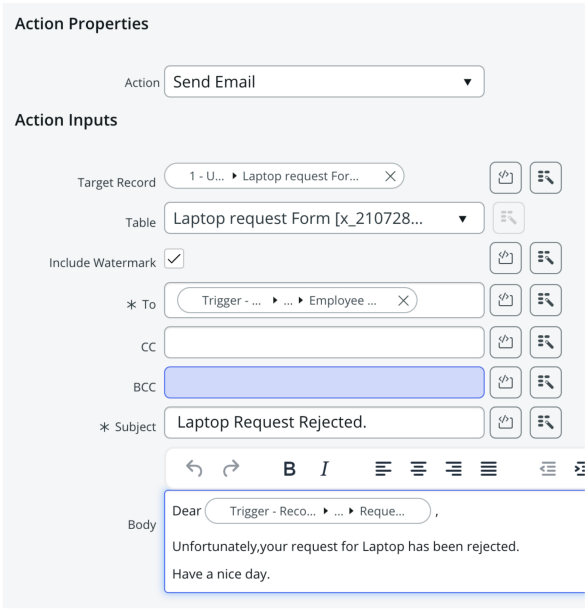


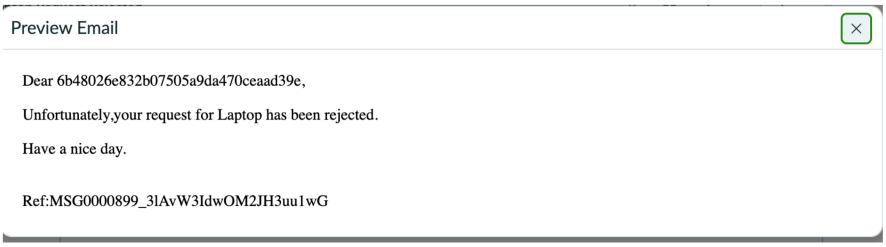
4. Requester approval email



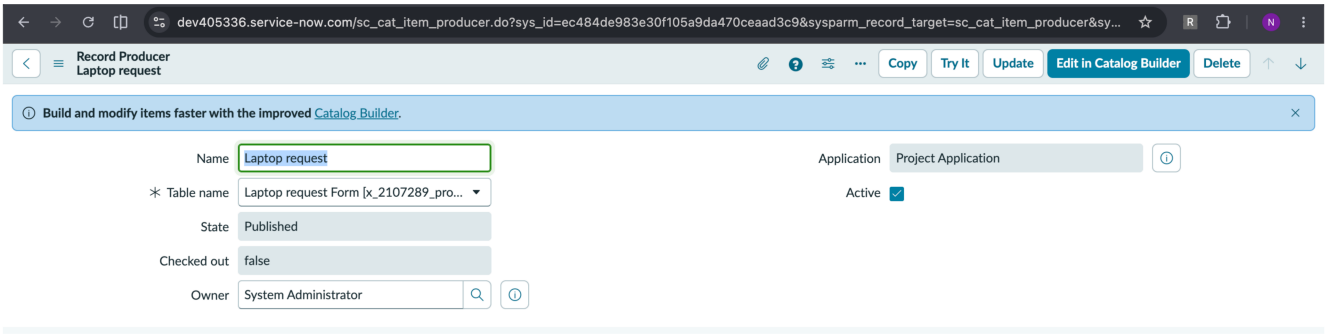


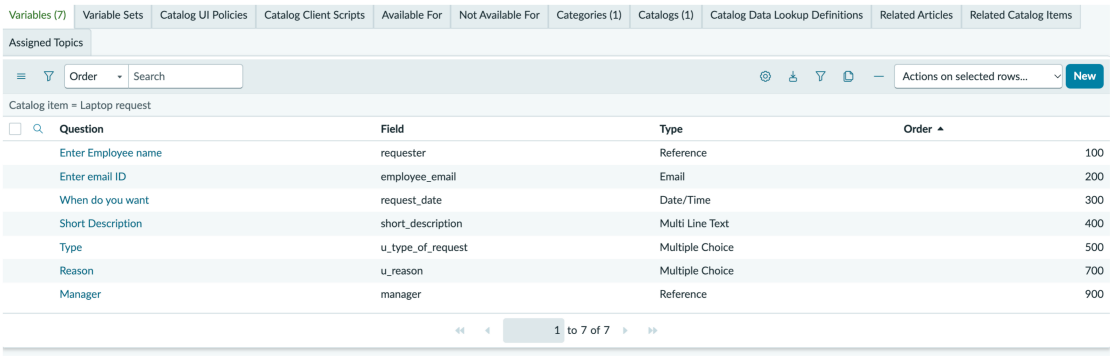
5. Requester rejection email



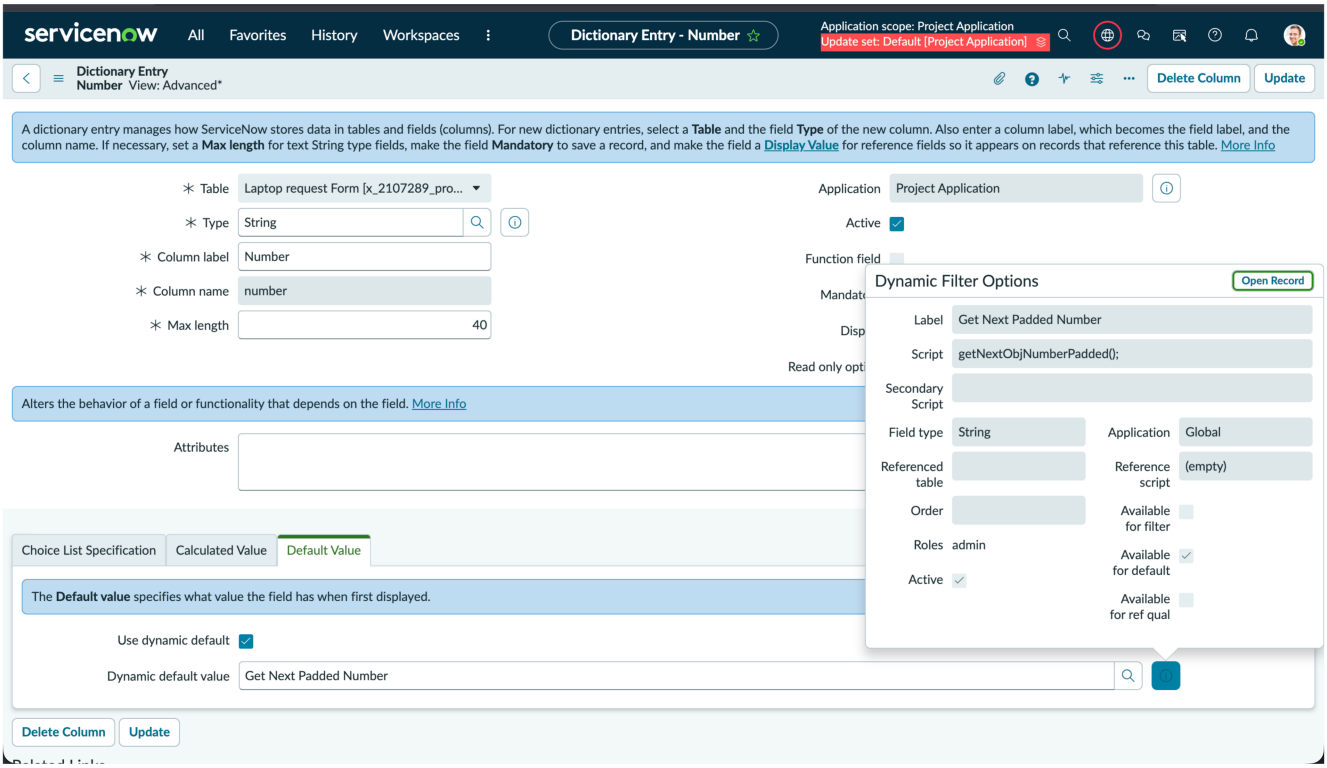


6. Service Portal Record Producer

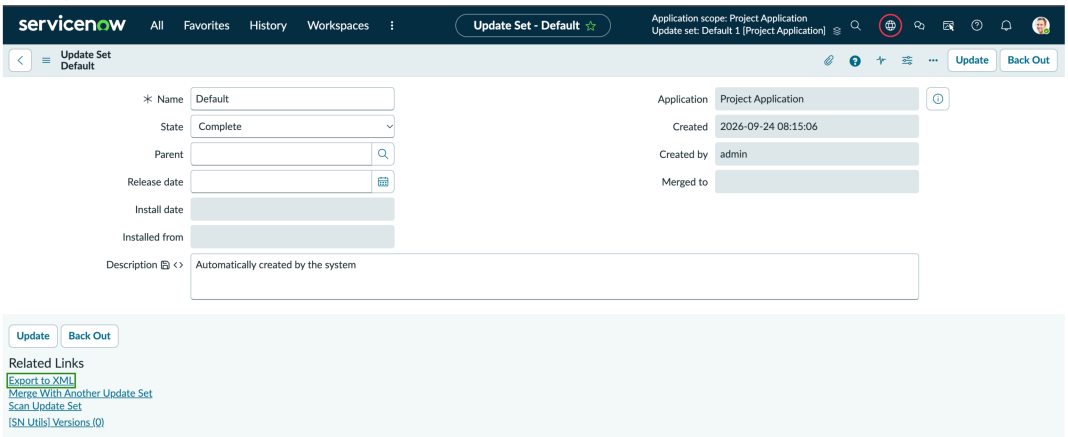




7. Request record with system-generated number

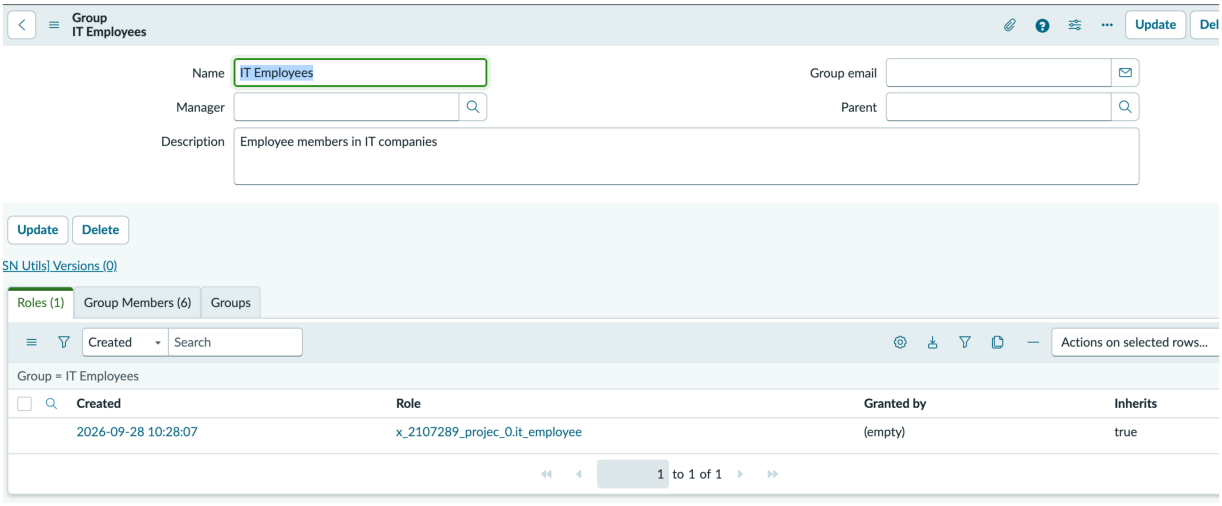


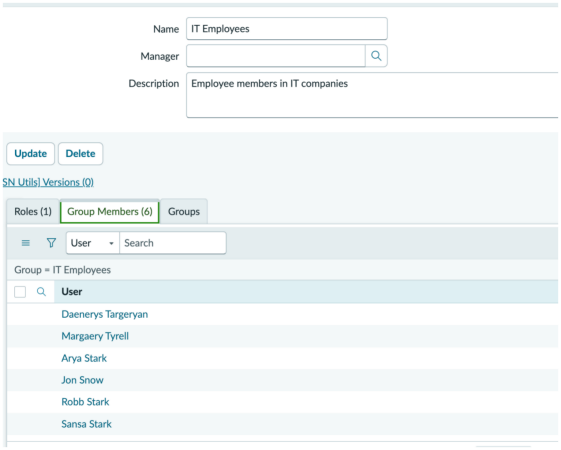
8. Update Set



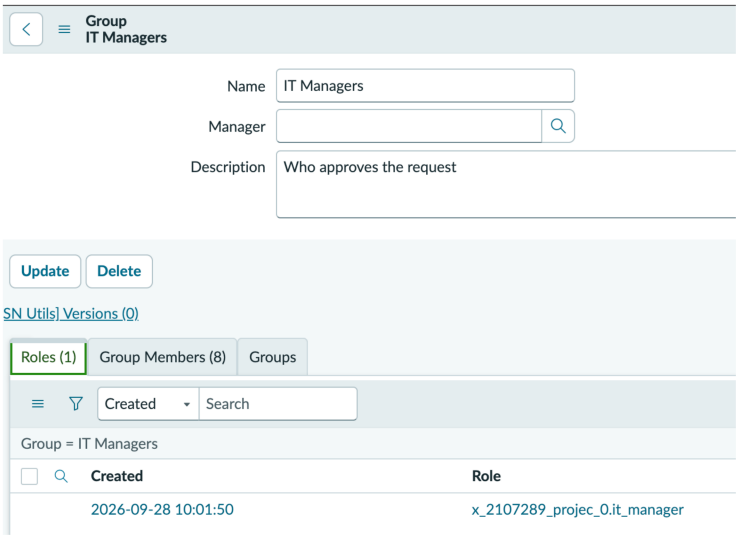
9. Groups and Roles

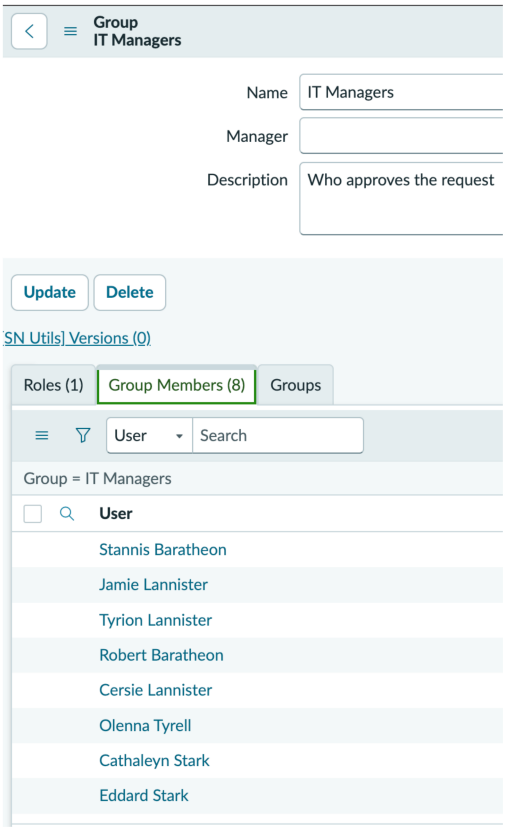
i) IT Employees



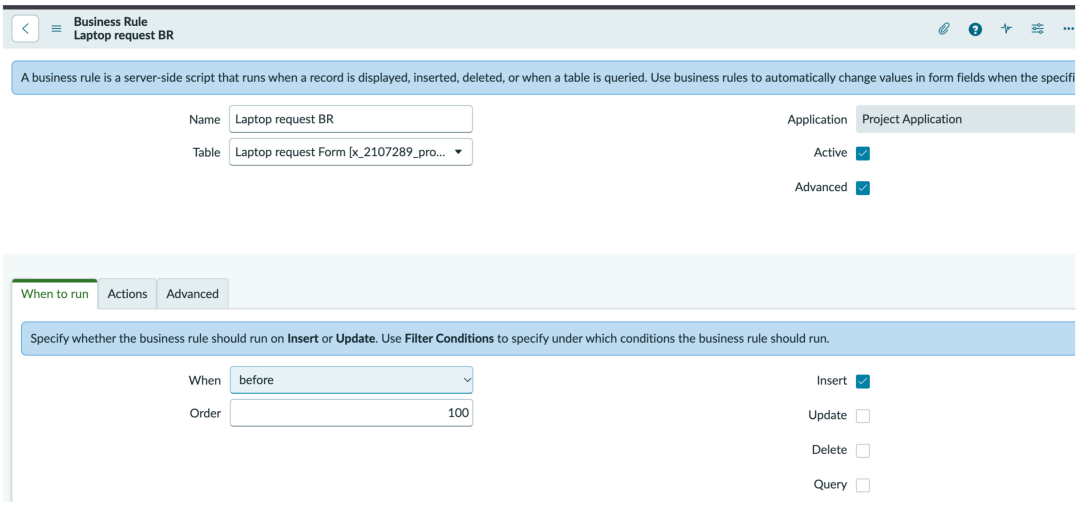


Ii) IT Managers





10. Business rules



When to run | Actions | Advanced

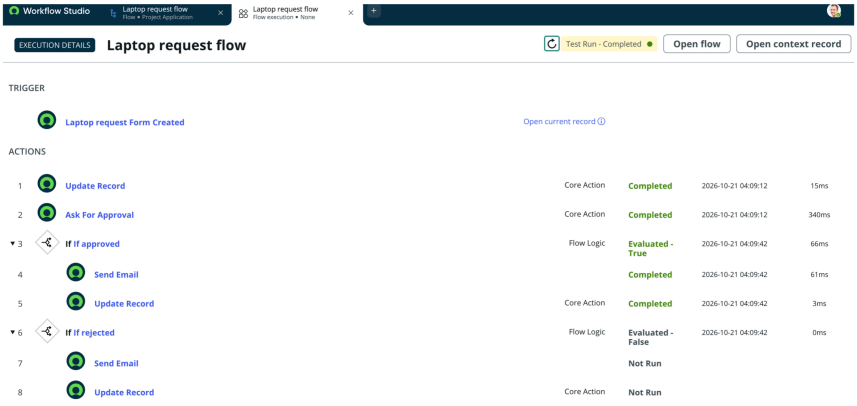
Condition

Script — Turn on ECMAScript 2021 (ES12) mode

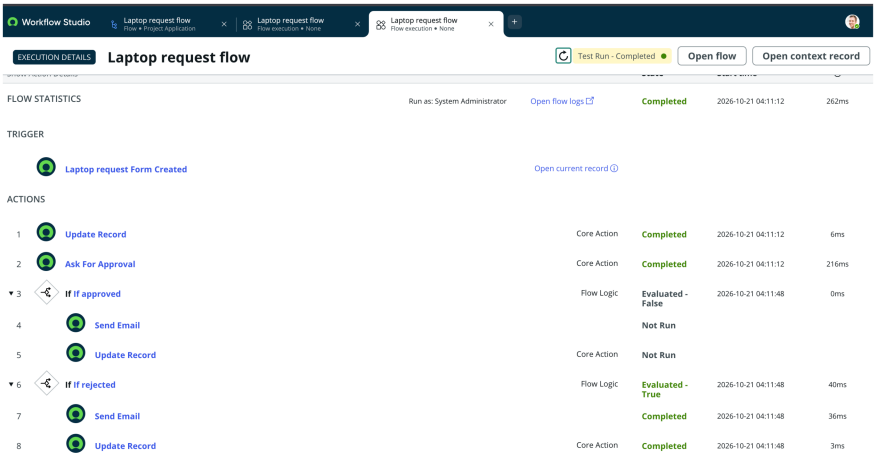
```
(function executeRule(current, previous) {

    var startOfYear = gs.beginningOfThisYear();
    var endOfYear = gs.endOfThisYear();

    // Check for existing request by same requester in the same year
    var gr = new GlideRecord('x_2107289_projec_0_laptop_request_form');
    gr.addQuery('requester', current.requester);
    gr.addQuery('sys_created_on', '>=', startOfYear);
    gr.addQuery('sys_created_on', '<=', endOfYear);
    gr.query();

    if (gr.next()) {
        gs.addErrorMessage('You have already submitted a laptop request for this year.');
        current.setAbortAction(true);
    }

})(current, previous);
```

11. Testing

I) Approved



Ii) Rejected



12. Known Issues

No additional issue is listed in this section. The final request-status transition and automatic IT task creation are treated as unverified behaviors, not as confirmed defects.

13. Future Enhancements

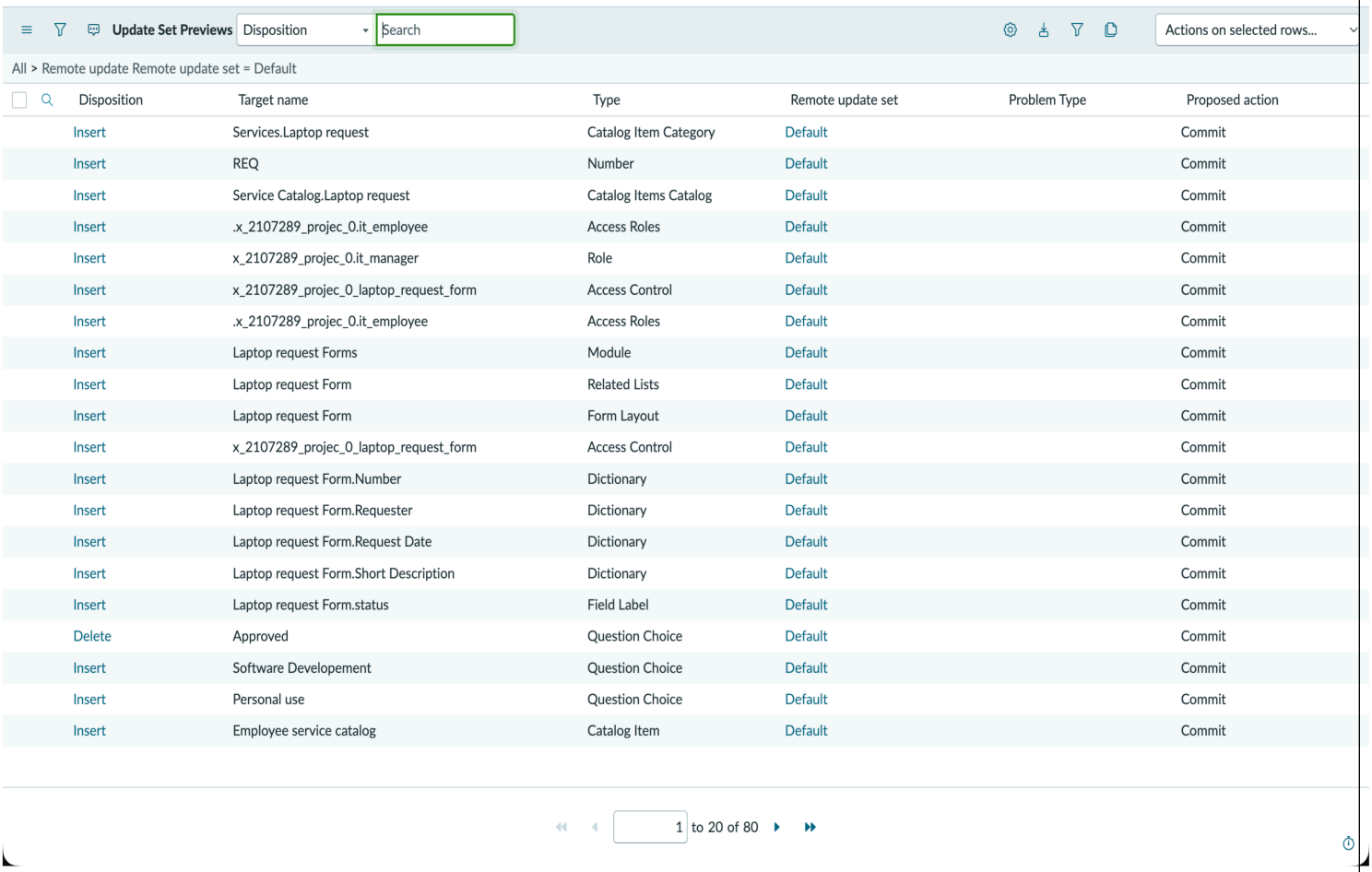
- Verify and, if required, implement automatic status transitions to Approved and Rejected after the approval decision.
- Add IT fulfillment task creation and assignment if it is required by the organization's process.
- Provide a clear request-tracking view for employees and IT staff, if needed.
- Improve reporting for request volume, approval turnaround time, and laptop type distribution.
- Expand regression testing for role-based access, ACLs, and approval outcomes.

Appendix A. Deployment Summary

The Update Set was exported as an XML file, imported into the target ServiceNow instance, previewed, conflicts were resolved, and the Update Set was committed. The project owner confirmed that the application is working in the target instance.

ServiceNow-Based Employee Laptop Request Application Update Set

Preview



Disposition	Target name	Type	Remote update set	Problem Type	Proposed action
Insert	Services.Laptop request	Catalog Item Category	Default		Commit
Insert	REQ	Number	Default		Commit
Insert	Service Catalog.Laptop request	Catalog Items Catalog	Default		Commit
Insert	.x_2107289_projec_0.it_employee	Access Roles	Default		Commit
Insert	x_2107289_projec_0.it_manager	Role	Default		Commit
Insert	x_2107289_projec_0_laptop_request_form	Access Control	Default		Commit
Insert	.x_2107289_projec_0.it_employee	Access Roles	Default		Commit
Insert	Laptop request Forms	Module	Default		Commit
Insert	Laptop request Form	Related Lists	Default		Commit
Insert	Laptop request Form	Form Layout	Default		Commit
Insert	x_2107289_projec_0_laptop_request_form	Access Control	Default		Commit
Insert	Laptop request Form.Number	Dictionary	Default		Commit
Insert	Laptop request Form.Requester	Dictionary	Default		Commit
Insert	Laptop request Form.Request Date	Dictionary	Default		Commit
Insert	Laptop request Form.Short Description	Dictionary	Default		Commit
Insert	Laptop request Form.status	Field Label	Default		Commit
Delete	Approved	Question Choice	Default		Commit
Insert	Software Developement	Question Choice	Default		Commit
Insert	Personal use	Question Choice	Default		Commit
Insert	Employee service catalog	Catalog Item	Default		Commit

Appendix B. Main Technical Identifiers

Item	Value
Application	Project Application
Custom table	u_laptop_request_form
Approval table	sysapproval_approver
Flow trigger	Record Created on Laptop Request Form
Initial status	Pending Approval
Employee group / role	IT Employees / it_employee
Manager group / role	IT Managers / it_manager